

PRINCE GULIA

princegulia170306@gmail.com | 8527875112 | Delhi, India

[prince-portfolio-xi.vercel.app](#) | [github.com/Prince-Gulia](#) | [linkedin.com/in/princegulia](#)

PROFILE

BCA student with hands-on experience building backend systems using Node.js, Express.js, PostgreSQL, Redis, and WebSockets. Built asynchronous processing systems, real-time applications, and RAG-based document systems, with experience deploying applications on Render, Vercel, and Supabase.

EDUCATION

Bachelor of Computer Applications (BCA) — Guru Gobind Singh Indraprastha University (IINTM, Janakpuri)

2024 – Present | New Delhi, India | CGPA: 9.52 / 10.0

12th (CBSE) — Spring Meadows Public School 2023 – 2024 | 84.8%

10th (CBSE) — Hari Krishna Public School 2021 – 2022 | 83.33%

PROJECTS

DocuMind — AI Document Assistant

Jul 2026 – Aug 2026

Node.js, Express, PostgreSQL, pgvector, BullMQ, Gemini API | [Live Demo](#) | [GitHub](#)

- Architected a RAG pipeline using the Gemini API for vector embeddings and pgvector for cosine similarity search across document chunks.
- Designed a semantic (paragraph-boundary) chunking strategy in place of naive fixed-length splitting, reducing context fragmentation across chunk boundaries.
- Diagnosed Gemini API rate-limit behavior under real document loads — identified that a 1500KB+ document producing 50+ chunks exhausted the API quota, and that an experimental version adding per-chunk summaries and hypothetical-question generation exhausted the same quota within 3–5 chunks — findings that directly shaped scope decisions for the production pipeline.
- Built an asynchronous PDF processing queue with BullMQ and ioredis so uploads are handled in the background without blocking API requests; tuned the connection config to prevent ioredis from timing out BullMQ's blocking commands under Render's hosted environment.
- Designed strict AI query guardrails to keep responses document-grounded and eliminate out-of-scope hallucination.

File Processing API

May 2026 – Jun 2026

Node.js, Express, BullMQ, Redis, Multer, Cloudinary, Sharp | [Live Demo](#) | [GitHub](#)

- Built an async file ingestion API supporting image and PDF uploads up to 10MB via Multer, with queue-based background processing using BullMQ and Upstash Redis.
- Integrated Sharp for server-side image transformation (resize, compress, format conversion) and Cloudinary for cloud storage and CDN delivery.
- Implemented job-status polling endpoints (pending / processing / completed) so clients can track processing state without blocking the main request thread.
- Configured exponential backoff retry logic for failed jobs so transient upload errors (network blips, Cloudinary timeouts) self-recover without manual intervention, keeping the queue resilient under Render's hosted environment constraints.

RealChat — Real-Time Chat Application

Mar 2026 – Apr 2026

Node.js, Express, Socket.io, PostgreSQL, JWT | [Live Demo](#) | [GitHub](#)

- Built a real-time chat backend with Socket.io handling persistent WebSocket connections authenticated via JWT, supporting room-based messaging with accurate user-room mapping.
- Engineered a PostgreSQL schema for message history with B-tree indexing on frequently queried columns to speed up chat history retrieval.
- Implemented secure room-based broadcasting to isolate chat events per room and prevent cross-room data leakage.
- Optimised payload structure and connection event handling for message delivery consistently under 1 second, verified across 3 concurrent client connections in the same room.

TECHNICAL SKILLS

Languages — JavaScript, C++, SQL, Python

Backend — Node.js, Express.js, RESTful API Design, WebSockets, Socket.io, JWT Authentication, Bcrypt

Databases — PostgreSQL, pgvector, Supabase, MySQL, SQLite

Async / Infra — Redis, BullMQ, Job Queues, ioredis

AI / ML — RAG Pipelines, Vector Embeddings, Semantic Chunking, Gemini API, Semantic Search

File Handling — Multer, Cloudinary, Sharp (Image Processing)

Deployment — Render, Vercel, Supabase (hosted PostgreSQL), Upstash

Tools — Git, GitHub, Postman, Linux

ACHIEVEMENTS & CERTIFICATIONS

- Achieved Joint Rank 1 in BCA Semester 2.
- Solved 180+ problems on [LeetCode](#).
- HackerRank Certification — REST API (Intermediate).